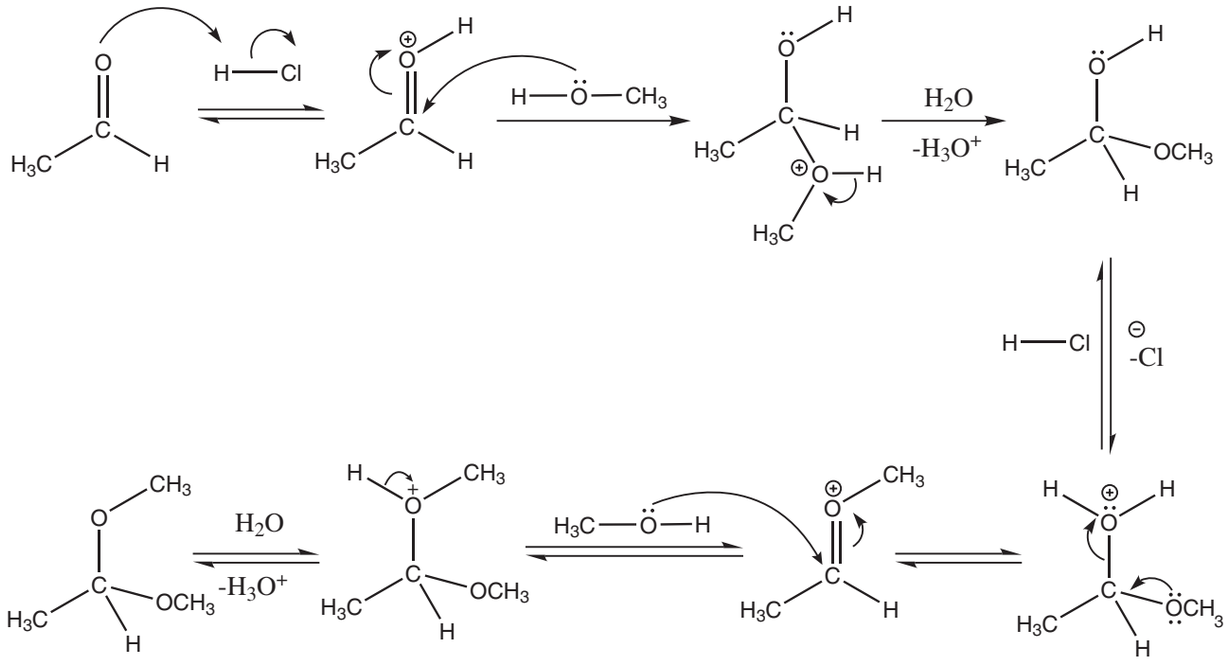
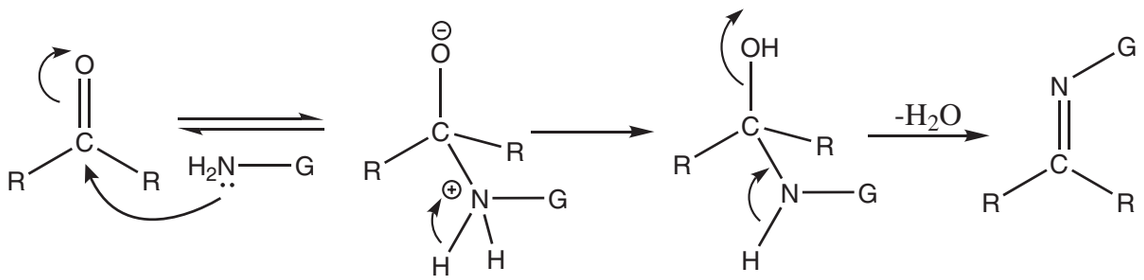
## Mechanisum



## 4) Addition of ammonia and its derivatives

When the nucleophiles, such as ammonia and its derivative $H_2\ddot{N}\text{-}G$ is treated with carbonyl compound, nuceophilic addition takes place, the carbonyl oxygen atom is protonated and then elimination takes place to form carbon – nitrogen double bond $\left(>C=N-G\right)$

When G – alkyl, aryl, OH, $NH_2$, $C_6H_5NH$, $NHCONH_2$ etc...



| G | Ammonia derivatives | Carbonyl derivatives | Product name |
|---|---|---|---|
| — OH | Hydroxyl amine | $>C=N-OH$ | Oxime |
| $-NH_2$ | Hydrazine | $>C=N-NH_2$ | Hydrazone |
| $-HN-C_6H_5$ | Phenyl hydrazine | $>C=N-NH-C_6H_5$ | Phenyl hydrazone |



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